The Dummy Variable Trap in Linear Regression

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Score	Age	Hours_Studied	IQ	Gender
65	18	2	105	Male
70	19	4	108	Female
72	20	3	110	Male
80	21	5	115	Female
85	22	6	112	Male

Male	Female
1	0
0	1
1	0
0	1
1	0

- $\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4D_1$
- Note: We discussed dummy variables earlier in (2- Encoding Categorical — One Hot Encoding)
  - You can find it here: https://github.com/Hazem1695/ml-concept-briefs
- Why don't you complete the equation to be like this?
  - $\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4D_1 + \boldsymbol{b_5D_2}$
- Let's discuss this trap.

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- Now notice:
  👉 Female = 1 – Male ($\boldsymbol{D_2} = 1 - D_1$)
- So:
  - 1. If you know Male → you automatically know Female
  - 2. This means perfect dependency
- ❌ The Problem in Linear Regression
  - 1. The model cannot uniquely determine coefficients
  - 2. Infinite solutions → unstable model

- ✅ The Solution (THIS is your key idea)
  - We drop one dummy variable
- Now:
  - If Male = 1 → Male
  - If Male = 0 → Female (implicitly)
- 👉 The dropped category is called:
  - Reference Category (Baseline)

- The dropped category is represented when all dummy variables = 0
- The intercept (b_0) becomes the baseline value

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Rules

- **Rule 1:**
  - If a column has 2 categories → use 1 dummy variable
- **Rule 2:**
  - If a column has k categories → use (k - 1) dummy variables
  - Example:

    20 categories → use 19
- **Rule 3**
  - If you have multiple categorical columns:
    Apply this rule to EACH column independently
  - Example:

    City (5 values) → 4 dummies

    Color (3 values) → 2 dummies

    Total = 6 dummy variables

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Final Clean Explanation

- Dummy Variable Trap happens when we include all dummy variables after encoding, which creates perfect multicollinearity.
- To avoid this, we always drop one dummy variable from each categorical feature.
- If a feature has k categories, we use (k - 1) dummy variables, and the dropped category becomes the reference (baseline).

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